Logic, Language, and Philosophy

Proposition logic: natural deduction (2)

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Characteristics

Derivations:

- Finite sequence of formulas
- Starts with premisses, ends with conclusion
- Steps:
  - assumption and marked as such; or
  - derived and marked with rule and steps used

Rules:

- Elimination rules
- Introduction rules
- Conditions on application of rules:
  - General: on the use of assumptions
  - Specific: on the use of e.g. Repetition rule

Rules so far

Rules we have introduced so far:

- Conjunction elimination and introduction
- Disjunction elimination and introduction
- Implication elimination and introduction
- Repetition

Note:

- Repetition: is not necessary (but convenient)
- Assumptions: need to be retracted in the inverse order in which they have been introduced

Negation

Negation elimination

Elimination rule $E_\neg$:

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m. & \neg\phi & \\
\vdots & & \\
k. & \phi & \\
\vdots & & \\
n. & \bot & E_\neg, m, k
\end{array}
$$

Here $\bot$ is a special atomic formula, called *Falsum*, which (like a contradiction) is always false

Example

$p, q \to \neg p, q \vdash \bot$

1.	p	$\mathtt{A}$
2	$q \to \neg p$	$\mathtt{A}$
3.	q	$\mathtt{A}$
4.	$\neg p$	$E_{\to}, 2, 3$
5.	$\bot$	$E_{\neg}, 1, 4$

Negation introduction

$$
\begin{array}{l}
\quad 1. \\
\quad \vdots \\
\left[\begin{array}{lll}
\quad m. & \phi & \text{ass.} \\
\quad \vdots & & \\
n-1 & \bot &
\end{array}\right. \\
\hline
\quad n. \quad \neg\phi \quad I_{\neg}
\end{array}
$$

Note: this treats $\neg\phi$ as $\phi \to \bot$

Example

$\vdash \neg(p \wedge \neg p)$

1.	$p \wedge \neg p$	A
2	p	$E_\wedge, 1$
3.	$\neg p$	$E_\wedge, 1$
4.	$\bot$	$E_\neg, 2, 3$
5.	$\neg(p \wedge \neg p)$	$I_\neg$

Minimal logic

Given by:

- $E_\neg$, $I_\neg$

Not derivable:

- $\neg\phi \to (\phi \to \psi)$
- $\phi \vee \neg\phi$

EFSQ-rule

EFSQ-rule ('Ex Falso Sequitur Quodlibet'):

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
n-1. & \bot & \\
n. & \phi & \text{EFSQ}, n-1
\end{array}
$$

Example

$p \vee q, \neg p \vdash q$

1.	$p \vee q$	A
2.	$\neg p$	A
3.	p	A
4.	$\bot$	$E_\neg, 2, 3$
5.	q	EFSQ, 4
6.	$p \to q$	$I_\to$
7.	q	A
8.	$q \to q$	$I_\to$
9.	q	$E_\vee, 1, 6, 8$

Intuitionistic logic

Given by:

- $E_\neg$, $I_\neg$, EFSQ

Not derivable:

- $\phi \vee \neg\phi$

Excursus: What's wrong with excluded middle?

Why would you not want excluded middle?

- Constructive reasoning
- Future indeterminism
- Vagueness
- Quantum phenomena
- ...

General characteristic:

- Excluded middle fails for phenomena that allow for 'truth value gaps'

Double negation

$\neg\neg$-rule:

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m. & \neg\neg\phi & \\
\vdots & & \\
n. & \phi & \neg\neg\text{-rule}, m
\end{array}
$$

Example

$q \to p, \neg q \to p \vdash p$

1.	$q \to p$		A
2.	$\neg q \to p$		A
3.	$\neg p$		A
4.	q	A	
5.	p	$E_\to, 1, 4$	
6.	$\bot$	$E_\neg, 3, 5$	
7.	$\neg q$	$I_\neg$	
8.	p	$E_\to, 2, 7$	
9.	$\bot$	$E_\neg, 3, 8$	
10.	$\neg\neg p$		$I_\neg$
11.	p		$\neg\neg$-rule, 10

Classical logic

Given by:

- $E_\neg$, $I_\neg$, EFSQ, $\neg\neg$-rule

Summary

	Minimal logic	Intuitionistic logic	Classical logic
Rules	$E_\neg$, $I_\neg$	$E_\neg$, $I_\neg$, EFSQ	$E_\neg$, $I_\neg$, EFSQ, $\neg\neg$-rule
Not derivable	$\neg\phi \to (\phi \to \psi)$, $\phi \vee \neg\phi$	$\phi \vee \neg\phi$	

Completeness

The summary suggests that *everything* is derivable in the natural deduction system for classical logic. But what does that mean?

Explication:

> *Everything that is valid according to the semantics of classical logic is derivable in the natural deduction system for it*

This is called completeness of classical propositional logic. More on that in meeting 25

Note: completeness is a *relation*, between a derivation system and a semantics

Now what about completeness of intuitionistic logic?

- We already have a derivation system for it
- But what about its semantics?

In meeting 27 we will introduce concepts that can be used to define it

Homework

Make exercises 8, 9 and 10.